Name ____________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 7
DUE: FRIDAY OCT. 31

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. *(12 points) Basic Modular Arithmetic*

(a) *(6 points)* Find $4^{-1} \mod 23$, if it exists. If it does not exist, exhaustively show why it can not exist.

$4x \equiv 1 \pmod{23}$

$4x - 1 = 23 \quad 4x = 24 \quad x = 6$

$\therefore \; 4^{-1} \bmod 23 = 6$

(b) *(6 points)* Find $4^{-1} \mod 26$ if it exists. If it does not exist, exhaustively show why it can not exist.

$\gcd(4,26) = 2 \quad 2 \neq 1$

$\therefore \; 4^{-1} \bmod 26 = \text{DNE}$

Problem 2. *(8 points) Basic Modular Arithmetic*
Suppose a and b are integers such that $a \equiv 4 \pmod{23}$ and $b \equiv 12 \pmod{23}$.

(a) *(4 points)* Find an integer $-46 \le c \le -24$ such that $c \equiv (b^2 - 4a) \pmod{23}$.

Let $a = 4, b = 12$

$(12^2 - 4a) \pmod{23}$

$144 - 16 \equiv 6 - 16 \pmod{23}$

find c such that $-46 \le c \le -24$
and $c \equiv -10 \pmod{23}$

$c \equiv 13 \pmod{23}$

$-33 \equiv 13 \pmod{23}$

$c \equiv -33$

(b) *(4 points)* Find an integer $23 \le d \le 45$ such that $a \cdot d \equiv b \pmod{23}$

Let $a = 4, b = 12$

$4d = 12 \pmod{23}$

$(4)(3) = 12$

to satisfy $23 \le d \le 45$, we add 23

$4(26) \equiv 12 \pmod{23}$

$(4)(26) \equiv 104 \equiv 12 \pmod{23}$

$d = 26$

Name ______________________________ Section ________

Problem 3. *(12 points) Basic Modular Arithmetic*
Evaluate the following expressions.

(a) *(6 points)* $\left(3^{99} + (-135) \cdot (35)\right) \mod 13$

$3^{99} = (3^3)^{33}$ $\quad 3^3 = 27 \quad 27 \bmod 13 = 1$ thus, $3^{99} \bmod 13 = 1$

$-135 \bmod 13 \equiv -5 \bmod 13 \equiv 8$

$35 \bmod 13 \equiv 9$

$(9)(8) = 72 \quad 72 \bmod 13 = 7$

$(1+7) \bmod 13 = 8$

thus, $(3^{99} + (-135)(35) \bmod 13 = (1 + (8)(9) \bmod 13 = (1+7) \bmod 13 = 8$

(b) *(6 points)* $\left((5^3 \mod 23)^2 \mod 31\right) \mod 4$

$5^3 \bmod 23 = ((25 \bmod 23)(5 \bmod 23)) \bmod 23 = ((2)(5) \bmod 23 = 10$

$10^2 \bmod 31 = 100 \bmod 31 = 7$

$7 \bmod 4 = 3$

$\therefore \quad ((5^3 \bmod 23)^2 \bmod 31) \bmod 4 = (10^2 \bmod 31) \bmod 4 = 7 \bmod 4 = 3$

Problem 4. *(12 points) Primes and Greatest Common Divisors*

(a) *(6 points)* Use Trial Division to determine whether 221 is prime or composite. Show every step.

$14 < \sqrt{221} < 15$

$2 \nmid 221$
$3 \nmid 221$
$4 = 2^2$, skip
$5 \nmid 221$
$6 = 3(2)$, skip
$7 \nmid 221$
$8 = 2^3$, skip
$9 = 3^2$, skip
$10 = 2(5)$, skip
$11 \nmid 221$
$12 = 2(6)$, skip
$13 \mid 221$

as $13 \mid 221$, 221 is composite

(b) *(6 points)* Use Trial Division to determine whether 211 is prime or composite. Show every step. only necessary to check primes

$14 < \sqrt{211} < 15$

$2 \nmid 211$
$3 \nmid 211$
$5 \nmid 211$
$7 \nmid 211$
$11 \nmid 211$
$13 \nmid 211$

thus, as there is no prime integer less than $\sqrt{211}$ that divides 211, 211 is prime

Problem 5. *(8 points) Primes and Greatest Common Divisors*
Find the prime factorization of the following integers.

(a) 504

$504 = (8)(63) = (2^3)(63) = (2^3)(9)(7) = (2^3)(3^2)(7)$

(b) $\frac{16!}{5!11!}$

$$\frac{(12)(13)(14)(15)(16)}{(1)(2)(3)(4)(5)} = \frac{\cancel{(3)}\cancel{(4)}(13)\cancel{(2)}(7)\cancel{(5)}(3)(2^4)}{\cancel{(3)}\cancel{(4)}\ \cancel{(2)}\ \cancel{(5)}\ (1)} = (13)(7)(3)(2^4)$$